

PURDUE SPACE PROGRAM: A SEDS Chapter

Condor Dome Regulators

PURDUE SPACE PROGRAM Team Name

Tescom 26-11xx Series Regulators

26-11	2	1	-	3	8	1

BASIC SERIES	BODY MATERIAL	LOADING METHOD	INLET AND OUTLET PORT TYPE	DOME PORT	INLET AND OUTLET PORT SIZE	INNER VALVE SIZE
26-11	**1 –** Brass **2 –** 303 Stainless Steel **3 –** Aluminum 2024-T351 **6 –** 316 Stainless Steel	**1 –** Externally loaded	**1 –** SAE **2 –** NPTF **3 –** MS33649	1/4" MS33649 1/4" NPTF 1/4" MS33649	**6 –** 3/8" **8 –** 1/2"	**1 –** 0.25" Orifice **2 –** 0.375" Orifice

OPERATING PARAMETERS

Pressure rating per criteria of ANSI/ASME B31.3

Maximum Inlet Pressure
Stainless Steel Body (0.25" Orifice): 10,000 psig / 690 bar
Stainless Steel Body (0.375" Orifice): 6000 psig / 414 bar
Aluminum Body: 6000 psig / 414 bar
Brass Body: 6000 psig / 414 bar

Outlet Pressure Ranges
To maximum inlet

Design Proof Pressure
150% maximum rated

Leakage
Bubble-tight

Operating Temperature
-40°F to 165°F / -40°C to 74°C

Flow Capacity
0.25" Orifice: $C_V = 0.46$
0.375" Orifice: $C_V = 1.30$

MEDIA CONTACT MATERIALS

Body
303 Stainless Steel, 316 Stainless Steel, Aluminum, or Brass

Diaphragm
Nitrile, Buna-N

O-Rings
Nitrile, Buna-N

Back-up Rings
PTFE

Seat
PCTFE

Retaining Ring
15-7 Stainless steel

Valve Cap
17-4 PH Stainless Steel

Remaining Parts
300 Series Stainless Steel

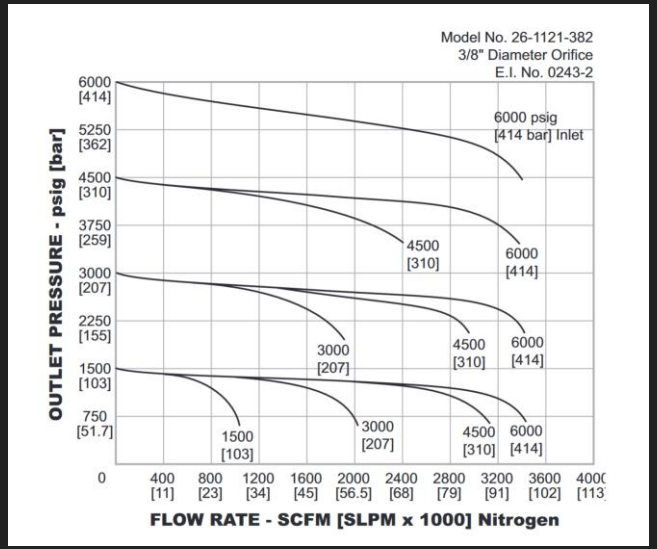
OTHER

Cleaning
CGA 4.1 and ASTM G93

Weight (without gauges)
Stainless Steel: 10 lbs / 4.5 kg
Aluminum: 4 lbs / 1.8 kg

Tescom 26-1111-382-158

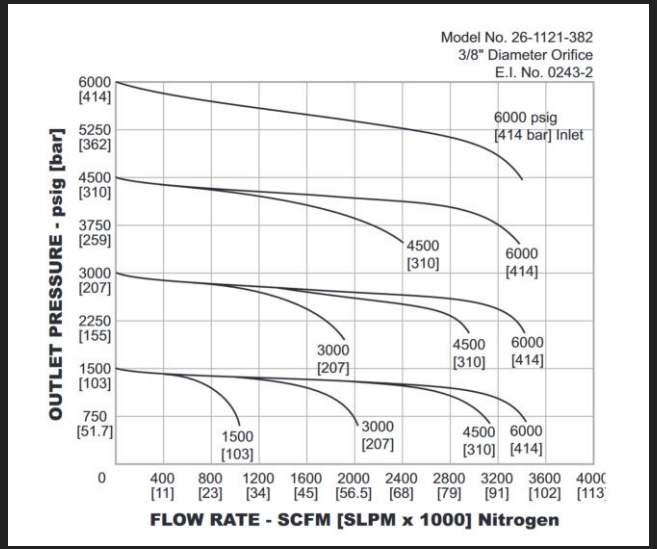
- Type: High-flow pressure-reducing dome-load regulator
- Current usage: DR-OX-017
- Max Inlet: 6000 psig
- Max Outlet: 6000 psig
- Cv: 1.3
- Material: Brass
- Inlet/outlet/load port type: MS33649
- Load port size: ¼”
- Port size: in question; serial number disagrees with provided
- Meets flow requirements for COPV and direct press
- KEEP





Tescom 26-1121-382

- Type: High-flow pressure-reducing dome-load regulator
- Current usage: DR-FU-005
- Max Inlet: 6000 psig
- Max Outlet: 6000 psig
- Cv: 1.3
- Material: Aluminum
- Inlet/outlet/load port type: MS33649
- Load port size: ¼”
- Port size: ½”
- Meets flow requirements for COPV and direct press
- KEEP





Tescom 26-12xx Series Regulators

$C_V = 3.3$

26-12 2 1 - 3 16 1

BASIC SERIES	BODY MATERIAL	LOADING METHOD	INLET AND OUTLET PORT TYPE	DOME PORT	PORT SIZE	ORIFICE SIZE
26-12	**2** – 303 Stainless Steel **6** – 316 Stainless Steel	**1** – External	**1** – SAE **2** – NPTF **3** – MS33649	1/4" MS33649 1/4" NPTF 1/4" MS33649	**12** – 3/4" **16** – 1"	**1** – 1/2" 12.7 mm

OPERATING PARAMETERS

Pressure rating per criteria of ANSI/ASME B31.3

Maximum Inlet Pressure
3600 and 6000 psig
248 and 414 bar

Outlet Pressure
To maximum inlet

Design Proof Pressure
150% maximum rated operating

Leakage
Bubble-tight

Flow Capacity
C_V = 3.3, 6.0, 12.0* or 20.0

MEDIA CONTACT MATERIALS

Body
303 Stainless Steel, 316 Stainless Steel

Seat
PCTFE or Polyimide (Vespel®)

Diaphragm
Nitrile, Buna-N or FKM (Viton®-A)

O-Rings
Nitrile, Buna-N or FKM (Viton®-A)

Back-up Rings
PTFE

Remaining Parts
300 Series Stainless Steel

Tescom 26-1211-3121-094

- Type: High-flow pressure-reducing dome-load regulator
- Current usage: SR-OX-207
- Max Inlet: 3000 psig
- Max Outlet: 3000 psig
- Cv: unknown
- Material:
- Inlet/outlet/load port type:
- Load port size:
- Port size:
- Does not meet pressure requirements, do not prioritize
- Note: spec sheet does not appear to match legacy regulator